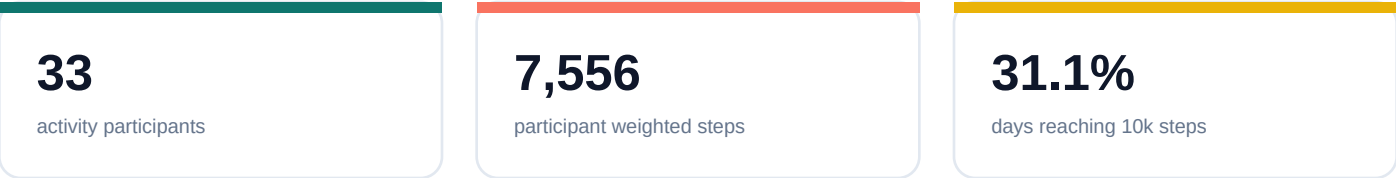


Bellabeat Usage Intelligence

Executive brief | Evidence led product analytics

Business question: What can this wearable usage sample support about activity patterns, product opportunities, and the evidence needed before rollout?

APR TO MAY
2016 SAMPLE



WHAT THE EVIDENCE SAYS

- 1

Saturday activity is highest
Participant weighted steps peak at 8,400 on Saturday. Hourly activity peaks near 18:00.
- 2

Sedentary time remains substantial
The participant weighted mean is 16.63 sedentary hours per recorded day, with a 95% interval from 15.38 to 17.86.
- 3

Sleep evidence is inconclusive
The within participant estimate is negative, but its cluster bootstrap interval includes zero. It does not establish an effect.

WHAT BELLABEAT SHOULD TEST

- 1

Evening movement prompt
Randomize a personalized prompt around the evening activity window. Measure incremental steps and prompt opt outs.
- 2

Weekly progress reflection
Test attainable, adaptive goals against a neutral summary. Measure goal completion and four week retention.
- 3

Sleep logging support
Test education and user controlled reminders. Measure valid sleep logs without increasing notification fatigue.

DECISION LIMITS

Coverage

33 activity, 24 sleep, and 8 weight participants. Missingness is not random.

Generalization

A small 2016 convenience sample cannot represent Bellabeat customers or current behavior.

Causality

Observational associations support hypotheses, not product impact claims. Use randomized tests.

Method Verified source checks in DuckDB, equal participant weighting, cluster bootstrap intervals, and explicit sensitivity analysis.

QUALITY GATE: PASS